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The neighbor-joining method: a new method for reconstructing phylogenetic trees

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Abstract

A new method called the neighbor-joining method is proposed for reconstructing phylogenetic trees from evolutionary distance data. The principle of this method is to find pairs of operational taxonomic units (OTUs [= neighbors]) that minimize the total branch length at each stage of clustering of OTUs starting with a starlike tree. The branch lengths as well as the topology of a parsimonious tree can quickly be obtained by using this method. Using computer simulation, we studied the efficiency of this method in obtaining the correct unrooted tree in comparison with that of five other tree-making methods: the unweighted pair group method of analysis, Farris's method, Sattath and Tversky's method, Li's method, and Tateno et al.'s modified Farris method. The new, neighbor-joining method and Sattath and Tversky's method are shown to be generally better than the other methods.

Real-World Impact & Application

Tracking Virus Mutations (COVID & Flu) and 23andMe Ancestry: How do scientists figure out which new variant of a virus is spreading, or how do ancestry tests trace your family tree back generations? Neighbor-Joining takes genetic code differences and builds an evolutionary family tree in seconds.